

Geometry: dS(closed)

Coordinates:

$$(t, \psi, \theta, \phi)$$

Metric Tensor ($g_{\mu\nu}$):

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & \alpha^2 \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & \alpha^2 \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & 0 & 0 & \alpha^2 \sin^2(\psi) \sin^2(\theta) \cosh^2\left(\frac{t}{\alpha}\right) \end{bmatrix}$$

Christoffel Symbols ($\Gamma^\lambda_{\mu\nu}$):

$$\begin{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \alpha \sinh\left(\frac{t}{\alpha}\right) \cosh\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & \alpha \sin^2(\psi) \sinh\left(\frac{t}{\alpha}\right) \cosh\left(\frac{t}{\alpha}\right) & 0 \\ 0 & 0 & 0 & \alpha \sin^2(\psi) \sin^2(\theta) \sinh\left(\frac{t}{\alpha}\right) \cosh\left(\frac{t}{\alpha}\right) \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix} \begin{bmatrix} 0 \\ \frac{\sinh\left(\frac{t}{\alpha}\right)}{\alpha \cosh\left(\frac{t}{\alpha}\right)} \\ 0 \\ 0 \end{bmatrix}$$

Riemann Curvature Tensor ($R^\lambda_{\mu\nu\rho}$):

$$\begin{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ -\cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & 0 & 0 & 0 \\ -\sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \frac{1}{\alpha^2} & 0 & 0 \\ -\frac{1}{\alpha^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & -\sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & \frac{1}{\alpha^2} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{\alpha^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 0 & \frac{1}{\alpha^2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{\alpha^2} & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\cosh^2\left(\frac{t}{\alpha}\right) \\ 0 & 0 & 0 & 0 \\ 0 & \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) \\ 0 & 0 & \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \end{bmatrix} \end{bmatrix}$$

Ricci Tensor ($R_{\mu\nu}$):

$$\begin{bmatrix} -\frac{3}{\alpha^2} & 0 & 0 & 0 \\ 0 & 3 \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & 3 \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & 0 & 0 & 3 \sin^2(\psi) \sin^2(\theta) \cosh^2\left(\frac{t}{\alpha}\right) \end{bmatrix}$$

Ricci Scalar (R):

$$\frac{12}{\alpha^2}$$

Einstein Tensor ($G_{\mu\nu}$):

$$\begin{bmatrix} \frac{3.0}{\alpha^2} & 0 & 0 & 0 \\ 0 & -3.0 \cosh^2\left(\frac{t}{\alpha}\right) & 0 & 0 \\ 0 & 0 & -3.0 \sin^2(\psi) \cosh^2\left(\frac{t}{\alpha}\right) & 0 \\ 0 & 0 & 0 & -3.0 \sin^2(\psi) \sin^2(\theta) \cosh^2\left(\frac{t}{\alpha}\right) \end{bmatrix}$$

Weyl Tensor ($C^\lambda_{\mu\nu\rho}$):

[illegible]